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Anomaly Detection in weekly SIRGAS GNSS Time Series: A TimesNet assessment in real data

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Abstract

Time series anomaly detection is essential in a range of fields because it allows for the identification of atypical patterns that differ from the expected behavior. Time series data of GNSS coordinates of SIRGAS stations are valuable for research purposes, including the determination of tectonic drift (velocity) and strain rate deformation. Changes in the patterns of a GNSS time series at a station due to abnormal events such as earthquakes, equipment changes, or malfunctions can significantly impact the accuracy of velocity estimation. In this work, we assess the efficacy of a neural network called TimesNet for detecting anomalies in real GNSS time series data from 95 SIRGAS stations in Brazil and Ecuador. As TimesNet achieved state-of-the-art efficiency on many benchmark datasets, it was chosen as the primary model for this research. Hyperparameter optimization was employed for TimesNet using two stations: BRAZ, in a low seismic activity area in Brazil, and CHEC in a high seismic activity area in Ecuador. The results with both hyperparameters sets were compared against three established anomaly detection filters (Gaussian process, Kalman and Moving Average) and a LSTM neural network that were paired with three anomaly scorer methods (norm, K-means and difference). We tested two methods to predict labels from the anomaly scores: 1) Using a fixed percentile (Top 1%) threshold, and 2) Using the Top S scores, where S is the number of anomalies of each station. The TimesNet with hyperparameters optimized in the CHEC station obtained better results than BRAZ, both in Brazil and Ecuador stations, showing that the choice of hyperparameters may present an impact in the anomaly detection results. The best $F1_{Top\ S}$ and MSE in Brazil were obtained with the Kalman filter ($F1_{Top\ S}$=57.60 | MSE = 0.0094), closely followed by TimesNet ($F1_{Top\ S}$=55.67 | MSE = 0.0095). In Ecuador, the best $F1_{Top\ S}$ and MSE were obtained by TimesNet ($F1_{Top\ S}$=63.64 | MSE = 0.0145), followed by the Kalman filter ($F1_{Top\ S}$=59.09 | MSE = 0.0344). Therefore, the anomaly scores produced by TimesNet were less noisy than those produced by the other methods on Ecuador. To foster further research, the source code for this analysis is publicly available at https://github.com/mauriciodev/deep_anomaly_gnss.

Introduction

Time series anomaly detection (AD) plays an important role in a variety of fields by identifying outliers and unusual patterns that deviate from an expected behavior in sequential data. Its use in economics (Braei & Wagner, 2020), information technology (IT) (Ji et al., 2021), healthcare (Schmidl et al., 2022) and geospatial fields (Barba et al., 2021a) could be used in the early detection of abnormal events, providing important information that could improve decision-making processes and the overall efficiency of many systems.

Within the geospatial field, AD can be used as a tool in climate change analysis (Choi et al., 2021), natural disaster prediction (Miau & Hung, 2020) and other areas, such as GNSS data monitoring (Barba et al., 2021b), with the latter being the focus of our work. AD in GNSS time series data could help to identify outliers and unusual patterns that could be related to abnormal events such as earthquakes and equipment changes/malfunctions (Ishikawa et al., 2020; Jing et al., 2022).

To date, many algorithms have been developed to perform AD on time series data. These algorithms can be divided into three different groups: statistical methods (SMs), machine learning (ML) and deep learning (DL) (Lim & Zohren, 2021). SMs include algorithms such as the Gaussian process (Williams & Rasmussen, 2006) and Moving Average algorithms. In the ML area, the well-established Kalman filter (Verhaegen & Verdult, 2007) is worth mentioning. Taking advantage of the nonlinearity provided by DL algorithms, one could mention long short-term memory (LSTM) networks (Lei et al., 2024), variational autoencoders (VAEs) (Desai et al., 2021) and transformer-based approaches (Ahmed et al., 2022).

Vitti (2012) used signal segmentation to detect anomalies in time series of GNSS coordinates, but the method could not handle missing data. Gazeaux et al. (2013) reported that most automated and semiautomated methods available at that time could not detect anomalies with lower magnitudes than manual inspection. Wu et al. (2017) designed software to perform anomaly detection in GNSS time series. The methods require hyperparameters that the user should interactively test. Several machine learning (Kiani, 2020) and time series decomposition (Ghaderpour, 2021; Lebrun et al., 2021) approaches have been developed for anomaly detection in time series of GNSS coordinates, but none have used state-of-the-art networks.

Recently, considering the difficulty of mapping deeply intricate temporal dependencies in time series data, Wu et al. (2022) presented a novel network architecture for performing anomaly detection in such data. TimesNet is a neural network that comprises the use of convolution neural networks and Fast Fourier Transform to map long-term temporal dependencies in time series data. Their architecture was tested against many benchmark datasets, achieving state-of-the-art (SOTA) performance.

Therefore, considering the results obtained by TimesNet in such datasets, we propose its use to perform AD on weekly SIRGAS GNSS time series data provided by DGFI-TUM (Zhang et al., 2024). We hypothesize that TimesNet could map the complex temporal patterns present in this kind of data, specifically, data from stations in Brazil and Ecuador. The Brazilian stations were selected for analysis because they exhibit consistent behavior related to their location in the middle of the continental South American plate, which is characterized by a low level of tectonic deformation. In contrast, Ecuadorian stations were employed because they frequently present abnormalities due to earthquakes caused by subduction at the boundaries between the Nazca Plate (oceanic) and the South American Plate.

To achieve our goal, we developed a Python code to download and preprocess weekly SIRGAS GNSS time series data from the selected stations. The preprocessed data were later used to train the TimesNet model with a set of different hyperparameters that were determined with the help of a hyperparameter optimization tool called the NNI (Yu & Zhu, 2020). The best models were subsequently analyzed to identify which hyperparameters play a decisive role in the overall performance of the model. The optimized parameters were used to detect anomalies at 77 SIRGAS stations in Brazil and 18 in Ecuador, which were assessed against other filtering models to verify the model efficiency.

Materials and Methods

In this section, we describe the most relevant aspects of the TimesNet architecture, the data preparation process, the details of the neural network training process for individual GNSS stations and the performance assessment process developed in our work.

TimesNet

TimesNet operates by dividing the time series into non-overlapping windows, where each window represents a specific time period. The network is trained on these windows to learn the patterns within the data. During inference, the network generates predictions for each window. The difference between these predictions to the actual values in the windows are output as anomaly scores, where large discrepancies indicate potential anomalies.

TimesNet is a deep neural network designed to model the temporal variation in 1D time series data. Real-world time series data usually present multiple periods, such as daily, weekly, and yearly, which interact with each other. Therefore, the network architecture captures temporal variations in a time series by decomposing 1D data in a 2D space that can explicitly map variations both within and between periods (Fig. 1).

This process depends on the discovery of periods present in the data. This is achieved by employing the Fast Fourier Transform (FFT). By analyzing the frequency domain of the FFT processed data, it is possible to verify that the first few peak frequencies are the most significant. Thus, to avoid noise originating from high frequencies, only the top-k frequencies are selected to determine which periods will be processed by TimesNet.

This leads to the overall modular architecture of TimesNet (Fig. 2), which is composed of a stack of so-called TimesBlocks that are connected in a residual way. A TimesBlock receives data in 1D space and processes these data with FFT to discover the most significant periods. Each period is then transformed into a 2D space and processed by a shared inception block that makes use of 2D convolutions to capture intra/interperiod variations. Each 2D space data point is then transformed back into 1D space data through an adaptive aggregation according to their relative importance, which is determined by applying SoftMax to the output of the 2D convolutions (Fig. 2).

The inception block used in the timesblock follows the first version published by Szegedy et al. (2014), which uses up to $2^{num_kernels + 1}$ parallel convolution layers with different kernel sizes (e.g.: 1x1, 3x3, 5x5), is responsible for capturing temporal variations in a 2D space (interperiod-variation and intraperiod-variation) as shown in the left part of Fig. 1. It can be replaced by various computer vision backbones to process 2D transformed space data. Visual backbones such as ResNet (He et al., 2016), ResNeXT (Xie et al., 2016), ConvNeXT (Liu et al., 2022) and attention-based models (Liu et al., 2021) can be explored, which brings flexibility to the network and takes advantage of the developments achieved by the computer vision community (H. Wu et al., 2022).

Since TimesNet was released as an open-source software, it was later adapted to some Python time series libraries. DeepOD is an open-source Python library that performs anomaly detection and outlier detection using deep learning methods, including TimesNet. The TimesNet implementation of DeepOD does not provide an early stop regularization technique. Therefore, considering the importance of such a technique, we developed an early stop function with patience and delta hyperparameters that were used in the training step of our experiments.

TimesNet Hyperparameters

The hyperparameters necessary to train TimesNet (Table 1) are those originally present in the DeepOD implementation plus the patience and delta that were included by our early stop implementation.

Table 1
TimesNet hyperparameters and default values

Name	Definition	Default Value
seq_len	Length of the input sequences	100
stride	Stride for sliding window on data	1
lr	Learning rate for optimizer	1e-4
epochs	Number of epochs to train the model	10
batch_size	Size of batches for training	32
epoch_steps	Number of steps per epoch	20
pred_len	Prediction length for the model	0
e_layers	Number of TimeBlocks to stack	2
d_model	Dimensionality of the model (number of features generated by the data embedding)	64
d_ff	Dimensionality of the feedforward layers (TimesBlock internal convolution dimension)	64
dropout	Dropout rate in the data embedding	0.1
top_k	Top K frequencies for FFT period finding	5
num_kernels	Number of convolutional layers in the Inception Block	6
patience	Number of epochs under convergence to early stop	3
delta	Convergence value for the early stopping	1e-7
percentile	Is not a model hyperparameter, but is used to make predictions	99%

Data preparation

SIRGAS records the GNSS observables of each continuously monitored station in the SIRGAS network. These observations are used to provide weekly solutions for the GNSS positions of each station monitored. A total of three files are made available for each station every week: NEU (north, east, and up coordinates), DSC (discontinuities) and XYZ (geocentric coordinates). The NEU local geodetic coordinates of a station represent the difference between the current week position and the first position in the time series (DN, DE, DU). The NEU file also provides the same information as the DSC file about the recorded events that could affect the time series, such as earthquakes and equipment changes/failures.

The discontinuities provided by SIRGAS do not represent every possible data problem in the time series. For example, some series may be affected by missing data, leading to unreported gaps in the DSC files, and certain minor equipment changes may not affect the station's position. Therefore, our preprocessing includes the edges of data gaps such as anomalies (Fig. 3).

For each selected station, the files are downloaded and preprocessed. The data extracted from the NEU file are used to create two CSV files that are used to train and evaluate TimesNet and the other comparison models, being the inputs used for each anomaly detection method. The first CSV file, made for training, contains the GPS week, DN, DE and DU. The second CSV file, used to assess model performance, contains the GPS week and a discontinuity label. If a recorded week contains a discontinuity, it is labeled 1; otherwise, it is labeled 0. The gaps were filled with the value 0. To represent the time series direction (from earlier to later), the first point in the gap and the first point after the gap were marked as anomaly labels.

Single-station hyperparameter optimization

TimesNet uses some hyperparameters that can be changed to adapt the network to different time series, thus defining different models for each set of hyperparameters. To fit a set of hyperparameters to the SIRGAS time series, we performed hyperparameter optimization for two stations. Considering that each station could be subject to different tectonic influences and different conditions regarding changes/failures of equipment, we performed the training steps individually. The objective was to determine a set of hyperparameters that could be shared among others to deliver similar performance metrics.

To determine the best hyperparameters, we used the Neural Network Intelligence (NNI) toolkit (Microsoft, 2021). We defined a search space for the TimesNet hyperparameters as described in Table 2.

Table 2
Hyperparameter search space for TimesNet.

HP	Search Space	Rational
seq_len	Choice in [10, 20]	Empirically determined: smaller values delivered better results for the GNSS SIRGAS stations
e_layers	Choice in [2,3]	Best results according to the original paper
d_model	Choice in [32, 64, 128]	Best results according to the original paper
d_ff	Choice in [32, 64, 128]	The original implementation usually uses d_ff = d_model
dropout	Quniform in [0.1, 0.3, 0.1]	Empirically determined: using the default value as starting point at steps of 5%
top_k	Choice in [3, 5]	Best result according to the original paper

In the training step, we obtained anomaly scores for each point in the time series. Two sets of predictions were calculated based on the anomaly scores: the top 1% (percentile 1) and the top S (where S is the number of anomalies in the truth set). The top 1% prediction set marks the top 1% scores as predicted anomalies, while the top S prediction set marks the top S scores. The latter was created to avoid the effects of selecting a percentile threshold.

During the test step, the truth set and each prediction set were used to calculate the accuracy, precision, recall, F1 and MSE metrics (Chicco & Jurman, 2020) to assess the performance of the model. Thus, we compared the unsupervised labels obtained from the anomaly scores against the labels determined during the preprocessing step (i.e. maintainer anomalies plus data gaps borders). The accuracy, precision, recall and F1 are calculated with the number of true positives (TPs), true negatives (TNs), false positives (FPs) and false negatives (FNs), and the MSE is calculated based on predictions and observations, as summarized in Table 3. We chose the MSE metric for the HPO search because it uses the anomaly scores output by the TimesNet decision function, not relying on predictions that are generated based on a percentile value.

Table 3
Metrics used and their definitions.

Metric	Description	Formula	Remarks
Accuracy	Determine the proportion of correct predictions over all samples	$acc = \frac{TP+TN}{TP+TN+FP+FN}$	Doesn't deal with class imbalance
Precision	Determine the proportion of predictions that are correct	$p = \frac{TP}{TP+FP}$	Disregards the number of FN that the model fails to identify
Recall	Determine the proportion of actual positives that were correctly predicted	$r = \frac{TP}{TP+FN}$	Disregards the number of FP that were incorrectly classified by the model
F1	Harmonic mean between Precision (p) and Recall (r)	$f1 = \frac{2(pr)}{p+r}$	Addresses limitations of precision and recall, providing a balanced assessment for imbalanced dataset
MSE	The mean-square error between observed and predicted values	$mse = \frac{1}{n}\sum_{i=1}^{n}(o_i - p_i)^2$	Larger errors are punished more heavily than smaller errors.

As the $F1_{topS}$ is calculated using the top S predicted labels, for each FP labeled there is a FN label, which makes FP equal to FN. This ultimately makes $Precision_{topS}$, $Recall_{topS}$ and $F1_{topS}$ equal. Therefore, we only present $F1_{topS}$ in the metrics table.

Multiple-station assessment

To assess the overall performance of TimesNet in detecting anomalies across all the selected stations (77 from Brazil and 18 from Ecuador), we developed a procedure to evaluate its general effectiveness.

The single-station training procedure was used to obtain optimized hyperparameters for two selected stations that presented visually distinct behaviors, notably, stations such as BRAZ and CHEC. Stations such as the BRAZ exhibit more consistent behavior, with smaller discontinuities normally originating from equipment changes or jumps in some of the positional components (i.e., jumps in N or E). Stations such as the CHEC present large discontinuities due to earthquakes, such as the 2016 Pedernales/Ecuador earthquake (Mw7.8) and the 2019 Palora/Ecuador earthquake (Mw7.5) registered at this station.

We evaluated the differences between the hyperparameters obtained with BRAZ and those obtained with CHEC. Therefore, the optimized hyperparameters obtained for each station were used in a global training process, which essentially included a single-station training procedure to obtain the predictions for each station used in our research. The predictions and labels of each station are then stacked to calculate the global metrics. TimesNet and the other comparison models produce an anomaly score for each point in

the input series as outputs from each model. Higher scores indicate a greater chance that the point is an anomaly. Building on the previously described performance metrics (Table 3), during the prediction step, anomaly scores in the top 1% percentile are labeled anomalies, while anomaly scores under the percentile parameter are labeled normal.

Comparisons with other models

We performed multiple-station assessments with other well-known models to compare the results obtained with those of TimesNet. We used the DARTS library (Herzen et al., 2022) and selected the following available anomaly detection filters: Kalman (Verhaegen & Verdult, 2007), Gaussian process (Williams & Rasmussen, 2006) and Moving Average. Additionally, we selected a LSTM neural network forecaster from DARTS library, using the hyperparameters that achieved the best results for GNSS coordinates time series forecast as stated by Wang et al. (2021). These models were paired with three anomaly scorer methods (e.g., Norm, K-means and Difference). Each filter was tested with each scorer method on every station in both datasets.

Results and discussion

Single-station hyperparameter optimization results

Our experiments started with a single-station training process with two stations, one from Brazil (BRAZ) and the other from Ecuador (CHEC). For each station, NNI was set to perform up to 50 experiments with different hyperparameter combinations drawn from the search space shown in Table 2. The hyperparameters with optimal performance after optimization were used to define two TimesNet models that were used in our experiments. The results are presented in Table 4.

Table 4
Hyperparameter optimization results.

Hyperparameters	BRAZ	CHEC
seq_len	10	10
e_layers	2	3
d_model	128	64
d_ff	32	128
dropout	0.3	0.4
top_k	4	5
MSE	0.0064	0.0140

The combination of the top hyperparameters (e.g., the hyperparameter set that achieved the best performance over all 50 experiments) generated by the HPO for BRAZ generated a TimesNet model with

two TimesBlocks that was able to reach a MSE of 0.0064.

Conversely, the top hyperparameter combination generated by the HPO made with CHEC generated a TimesNet model with three TimesBlocks that was able to reach a MSE of 0.0140. It is worth highlighting that for CHEC, the model demanded a d_ff hyperparameter four times larger than BRAZ to achieve its best result.

Multiple-station assessment results

To assess the effect of using either hyperparameter set found in the single-station experiment to perform anomaly detection in GNSS time series data, we executed multiple-station assessments at all selected stations in Brazil and Ecuador separately, exchanging the optimized hyperparameters obtained for BRAZ and CHEC. Table 5 contains the metrics obtained with both hyperparameter sets (trained on CHEC and BRAZ) using both datasets (Brazil and Ecuador).

Leveraging the best hyperparameters, we trained two TimesNet models to generate anomaly predictions for each station. TimesNet produces an anomaly score for each point in the input series. Higher scores indicate a greater chance that the point is an anomaly. During the prediction step, anomaly scores in the top 1% percentile are labeled anomalies, while anomaly scores under the percentile parameter are labeled normal.

Table 5
Global metrics calculated with exchanged hyperparameters with full processing time (training + inference).

HPO station	**Brazil**		**Ecuador**	
	BRAZ	**CHEC**	**BRAZ**	**CHEC**
Accuracy (%)	98.71	**98.81**	**98.08**	97.99
Precision (%)	55.96	**60.63**	**73.47**	69.39
Recall (%)	42.06	**45.58**	**32.73**	30.91
$F1_{1\%}$ (%)	48.03	**52.04**	**45.28**	42.77
$F1_{topS}$ (%)	51.47	**55.67**	62.73	**63.64**
MSE	0.0099	**0.0095**	**0.0145**	**0.0145**
Time (s)	**822.55**	1467.57	**90.97**	197.04

The differences between the results obtained with both hyperparameter sets were not greater than 5%. The precision, recall and $F1_{1\%}$ metrics were greater in Brazil than in Ecuador, with a greater percentage of anomalies detected with anomaly scores in the top 1% quantile. The $F1_{topS}$ metric provides a percentage of how many discontinuities each model found compared to the number of discontinuities in the labeled data. In the Brazil dataset, more than 55% of the discontinuities were found by the networks, while more

than 63% were found in the Ecuador dataset. This means that TimesNet was able to detect most of the anomalies at the Ecuadorian stations but misclassified higher anomaly scores (recall 45.58% in Brazil and 32.73% in Ecuador). This is probably because effects of an earthquake in the weekly time series might not occur on the first week after the event, so the method produced labels on the part of the time series where the coordinates began to change. This could be improved with better models for the earthquake effects for labeled data.

Comparisons with other models

Multiple-station results with DARTS methods are presented along with TimesNet results to aid comparisons. Table 6 presents the results of multiple-station processing for the 77 stations in Brazil. It is possible to observe that TimesNet obtained the second place in all the metrics, remaining only behind Kalman with the Norm scorer.

Table 6
Brazilian station results: Metric values for Gaussian, Kalman, Moving Average and LSTM with Norm, K-means and Difference scores side by side with TimesNet with full processing time (training + inference).

Experiment	Accuracy	Precision	Recall	$F1_{1\%}$	$F1_{topS}$	MSE	Time (s):
Gaussian process Norm	98.24%	34.24%	25.74%	29.39%	33.79%	0.0233	36.81
Gaussian process Kmeans	97.81%	13.68%	10.20%	11.69%	14.40%	0.0776	42.12
Gaussian process Diff	97.79%	13.12%	9.86%	11.26%	14.51%	0.2988	39.13
Kalman filter Norm	**98.84%**	**62.14%**	**46.71%**	**53.33%**	**57.60%**	**0.0094**	**35.39**
Kalman filter Kmeans	97.69%	7.75%	5.78%	6.62%	10.77%	0.0675	34.35
Kalman filter Diff	98.05%	25.34%	19.05%	21.75%	22.34%	0.3922	33.79
Moving avg Norm	98.51%	46.76%	35.15%	40.13%	43.88%	0.0119	20.64
Moving avg Kmeans	97.70%	8.77%	6.58%	7.52%	10.20%	0.0593	20.04
Moving avg Diff	97.90%	18.10%	13.61%	15.53%	16.33%	0.4821	20.04
LSTM forecaster Norm	98.76%	55.00%	43.89%	48.82%	51.87%	0.0897	219.83
LSTM forecaster Kmeans	98.64%	49.24%	39.30%	43.71%	45.47%	0.0520	220
LSTM forecaster Diff	97.77%	9.09%	7.26%	8.07%	8.46%	0.3597	220.11
Timesnet BRAZ parameters	98.71%	55.96%	42.06%	48.03%	51.47%	0.0099	822.55
Timesnet CHEC parameters	**98.81%**	**60.63%**	**45.58%**	**52.04%**	**55.67%**	**0.0095**	**1467.57**

Table 7 presents the results of multiple station processing for the 18 stations in Ecuador. It is also possible to observe that the same situation presented in Table 6 persists. TimesNet maintains second place in all the metrics, staying only behind Kalman with Norm scorer. The LSTM forecaster with the Norm scorer, which achieved the third best results, obtained significantly lower precision, recall and $F1_{1\%}$ than did TimesNet with CHEC parameters.

Table 7
Ecuador station results: Metric values for the Gaussian, Kalman, Moving Average and LSTM models with Norm, K-means and Difference scores side by side with TimesNet with full processing time (training + inference).

Experiment	Accuracy	Precision	Recall	$F1_{1\%}$	$F1_{topS}$	MSE	Time (s):
Gaussian process Norm	97.37%	40.82%	18.18%	25.16%	35.45%	0.0355	7.46
Gaussian process Kmeans	96.77%	12.37%	5.45%	7.57%	12.73%	0.0992	7.33
Gaussian process Diff	97.00%	23.47%	10.45%	14.47%	19.55%	0.2792	7.74
Kalman filter Norm	**98.17%**	**77.55%**	**34.55%**	**47.80%**	**59.09%**	**0.0344**	**6.7**
Kalman filter Kmeans	96.78%	13.27%	5.91%	8.18%	9.09%	0.1072	6.67
Kalman filter Diff	97.15%	30.61%	13.64%	18.87%	17.27%	0.3390	6.69
Moving average Norm	97.62%	52.04%	23.18%	32.08%	48.18%	0.0202	4.79
Moving average Kmeans	96.62%	6.12%	2.73%	3.77%	15.45%	0.0906	4.57
Moving average Diff	97.18%	31.63%	14.09%	19.50%	22.27%	0.3698	4.63
LSTM forecaster Norm	97.88%	65.31%	29.36%	40.51%	45.41%	0.1683	35.08
LSTM forecaster Kmeans	97.61%	53.06%	23.85%	32.91%	40.37%	0.1417	35.57
LSTM forecaster Diff	96.78%	15.31%	6.88%	9.49%	8.26%	0.3537	34.6
Timesnet BRAZ parameters	**98.08%**	**73.47%**	**32.73%**	**45.28%**	62.73%	**0.0145**	**90.97**
Timesnet CHEC parameters	97.99%	69.39%	30.91%	42.77%	**63.64%**	**0.0145**	197.04

In both Ecuador and Brazil datasets, we found that the Norm scorer obtained the best results with every filter (Gaussian, Kalman, Moving Average) and LSTM, while the K-means scorer obtained the worst results in every experiment. The improvement from Difference scorer to Norm scorer ranged from 64–242%. Future work could focus only on the Norm scorer because the other scorers cannot achieve similar results in these time series.

Using the best results from TimesNet and DARTS, we calculated the differences in the computed metrics for both sets of stations. Table 8 shows, side by side, the best results from TimesNet and DARTS, where it is possible to observe the metric differences.

Table 8
Comparison between the Kalman model with the Norm scorer and TimesNet with full processing time (training + inference).

Metric	Ecuador		Difference	Brazil		Difference
	Kalman/Norm	TimesNet (BRAZ)	(Kalman-TimesNet)	Kalman/Norm	TimesNet (CHEC)	(Kalman-TimesNet)
Accuracy (%)	**98.17**	98.08	0.09	**98.84**	98.81	0.03
Precision (%)	**77.55**	73.47	4.08	**62.14**	60.63	1.51
Recall (%)	**34.55**	32.73	1.82	**46.71**	45.58	1.13
$F1_{1\%}$ (%)	**47.80**	45.28	2.52	**53.33**	52.04	1.29
$F1_{topS}$ (%)	59.09	**62.73**	-3.64	**57.60**	55.67	1.93
MSE	0.0344	**0.0145**	0.02	**0.0094**	0.0095	-0.0001
Time (s)	**6.70**	90.97	-84.27	**35.39**	1467.57	-1432.18

For Ecuador, it is possible to observe that the absolute difference stays under 5% for all percentage metrics. For Brazil, it is possible to observe that the absolute difference stays under 2% for all percentage metrics. Similarly, the absolute MSE difference is 0.0001 for Brazil and 0.02 for Ecuador. The difference in MSE shows that the anomaly scores produced by TimesNet are more accurate, with peaks in the labels and values near zero in points that are not predicted as anomalies.

It is possible to observe that the differences are lower for the Brazilian stations, which tend to present more consistent behavior. Conversely, for the stations in Ecuador that tend to experience larger discontinuities due to earthquakes, the differences are greater.

Additionally, considering that Brazil has 77 stations and Ecuador has 18 stations, it is possible that the differences between TimesNet and the Kalman/Norm implementation for DARTS could decrease as the number of stations used in the multiple-station assessment increases.

Visual assessment of the results

The outputs of the training process at BRAZ and CHEC stations were plotted to aid in the visual assessment of model performance. The predictions and scores output by TimesNet are overlaid on the time series data and the registered discontinuities to provide a combined view of the input data and the predicted results. The plots depict registered discontinuities as vertical black dotted lines for reference, and the predicted anomaly labels are represented by red vertical lines. Figure 4 shows the plot for the BRAZ station.

For BRAZ, TimesNet was able to accurately detect most of the anomalous regions present in the time series. It is possible to see score spikes in the regions that are related to antenna and receiver changes registered in this station's NEU file. On the other hand, the network did not present score peaks for the first three anomalies, which can indicate that those anomalies did not generate sufficiently strong deviations from the expected pattern for this time series.

For the CHEC station, the resulting plot (Fig. 7) shows that TimesNet correctly presented score spikes in the regions related to the 2016 Pedernales/Ecuador earthquake (Mw7.8) and the 2019 Palora/Ecuador earthquake (Mw7.5) in the regions related to antenna change and in the data gaps. Most of the spikes were not selected as anomalies due to the 99% percentile value.

To better assess the performance of TimesNet, we performed AD on the same stations using Kalman/Norm scorer because it outperformed TimesNet. The results can be found in Fig. 6 and Fig. 7.

The anomaly scores provided by the Kalman filter with Norm scorer increase after the anomaly score peaks. This could be an issue because only the peaks are anomalies, and the points near them might be labeled anomalies depending on the quantile percentage chosen. Additionally, Kalman presented score spikes in two out of the first four anomalies (i.e., the 2nd and 3rd anomalies referring to antenna changes) that were not selected as anomalies due to the threshold value.

Unlike the anomaly scores obtained at the BRAZ station, the Kalman filter with the Norm scorer shows different patterns at the CHEC station. The anomaly scores increase until the data gap occurs near week 1900. Many points that are not anomalous received have anomaly scores above average before week 1900, reaching values higher than those of the previously labeled anomalies. This is undesirable because a higher threshold would incorrectly label points as anomalies. TimesNet presented a narrower distribution near the peaks, which suggests that it would be more accurate with higher percentile thresholds. The first four labels in CHEC illustrate why TimesNet obtained a higher $F1_{topS}$ (63.64%) than did Kalman/Norm (59.09%) at Ecuador stations, even though TimesNet achieved a worse $F1_{1\%}$ score (45.28% and 47.80%, respectively).

Conclusion

In this paper, we presented a complete process to download and preprocess real SIRGAS GNSS time series data that were later used as datasets in our anomaly detection implementation. We presented the concepts and overall architecture of TimesNet and compared the results obtained from it against three filtering models (Gaussian, Kalman and Moving Average) and a LSTM forecaster. We concluded that even though TimesNet obtained similar or better anomaly scores (measured by the MSE and $F1_{topS}$) on real data, the classification results with the 1% percentile threshold were lower than Kalman/Norm scorer implementation provided by DARTS, achieving second place among the tested models.

A hyperparameter optimization tool was used to tune TimesNet to achieve nearly optimal performance at two stations, according to our experiments. These optimized hyperparameters were used to train

TimesNet against all selected stations (77 from Brazil and 18 from Ecuador). The combined results were assessed against the DARTS filtering models, which showed that TimesNet was outperformed by Kalman filter with the Norm scorer implementation provided by DARTS by a considerably small margin.

Both models achieved good results on real data. Additionally, it was possible to visually assess that the anomaly scores output from TimesNet presented peaks at registered anomalies that were not selected as predictions due to the threshold value used (99%). TimesNet obtained the best $F1_{Top\ S}$ score in the Ecuador dataset, so it obtained the best results in series with more anomalous patterns. The lower MSE and visual inspection of the anomaly scores in the graphs suggests that TimesNet scores tend to fluctuate less than Kalman scores, which can make it less affected by the threshold value.

Both models presented peaks at the data gaps, showing that the decision to create anomalies in such gaps was correct according to our experimental analysis. This measure was able to improve model performance for all tested solutions. Future work could improve the procedure that assigns labels for earthquakes, providing a better model for when the coordinates begin to change.

In terms of processing time, TimesNet required considerably more processing time (i.e., hundreds of seconds) than did Kalman, which was able to process stations from both Brazil and Ecuador in a few minutes. We suggest starting future work by comparing the Kalman filter as the base model, as it is fast and achieves the best metrics in our datasets. It would be beneficial to include globally distributed stations in future research, considering our focus on stations in Brazil and Ecuador. Additionally, we recommend using the procedure presented here with various time series, including those generated daily.

Considering that our work was performed only at stations in Brazil and Ecuador, we propose, as future work, the use of stations from other countries. We also propose the use of the procedure presented here with different time series, such as those provided by the NGL.

Finally, we conclude that our preprocessing algorithm can be used to create real-world benchmark datasets for anomaly detection in GNSS data. Additionally, to foster future research, we offer a GitHub page with our scripts that can be promptly used at any SIRGAS station, demanding the creation of a text file containing the names of the stations to be processed.

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Declarations

Ethics approval and consent to participate

Not applicable.

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Author Contribution

All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by L.C.O. de Andrade and M.C.M. de Paulo. The first draft of the manuscript was written by L.C.O. de Andrade and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data Availability

Code and data were released in the GitHub address https://github.com/mauriciodev/deep_anomaly_gnss to provide repeatability and reproducibility, while also providing new tools for the geodesy researchers.

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Figures

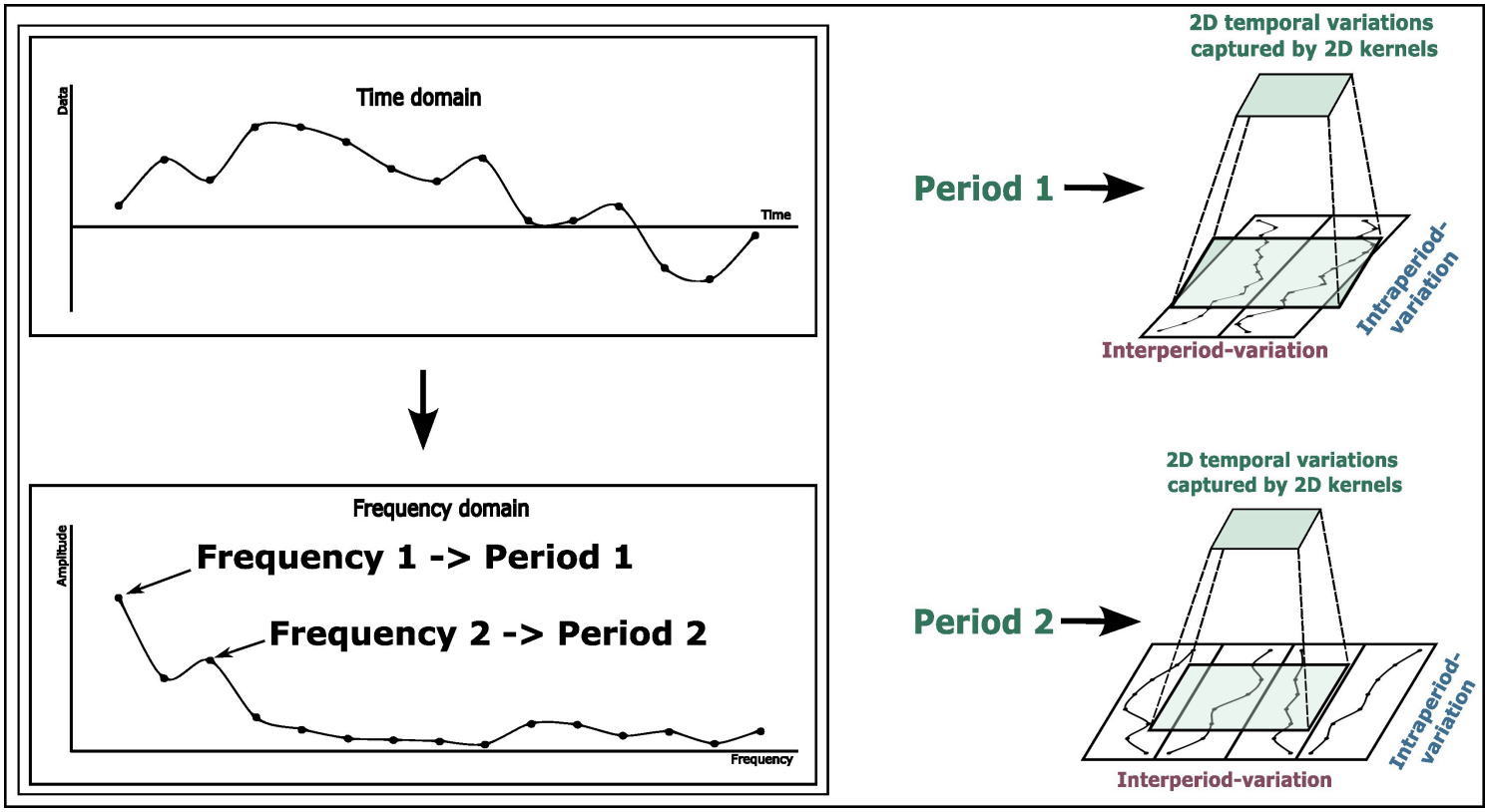


Figure 1

Main idea behind TimesNet: Represent 1D data in a 2D space to capture intra/interperiod temporal variations with the use of convolutions.

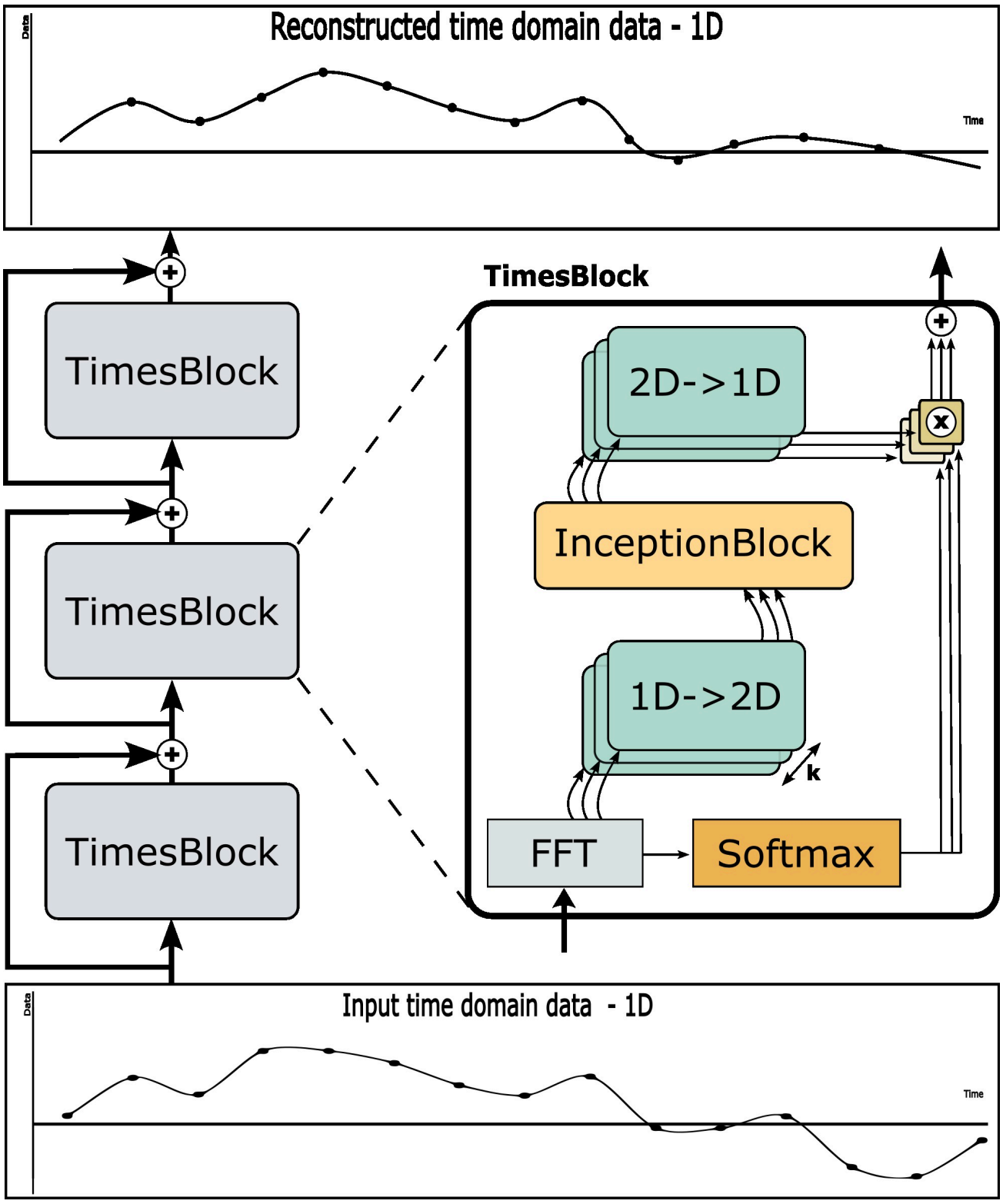


Figure 2

Overall TimesNet architecture (Wu et al., 2022).

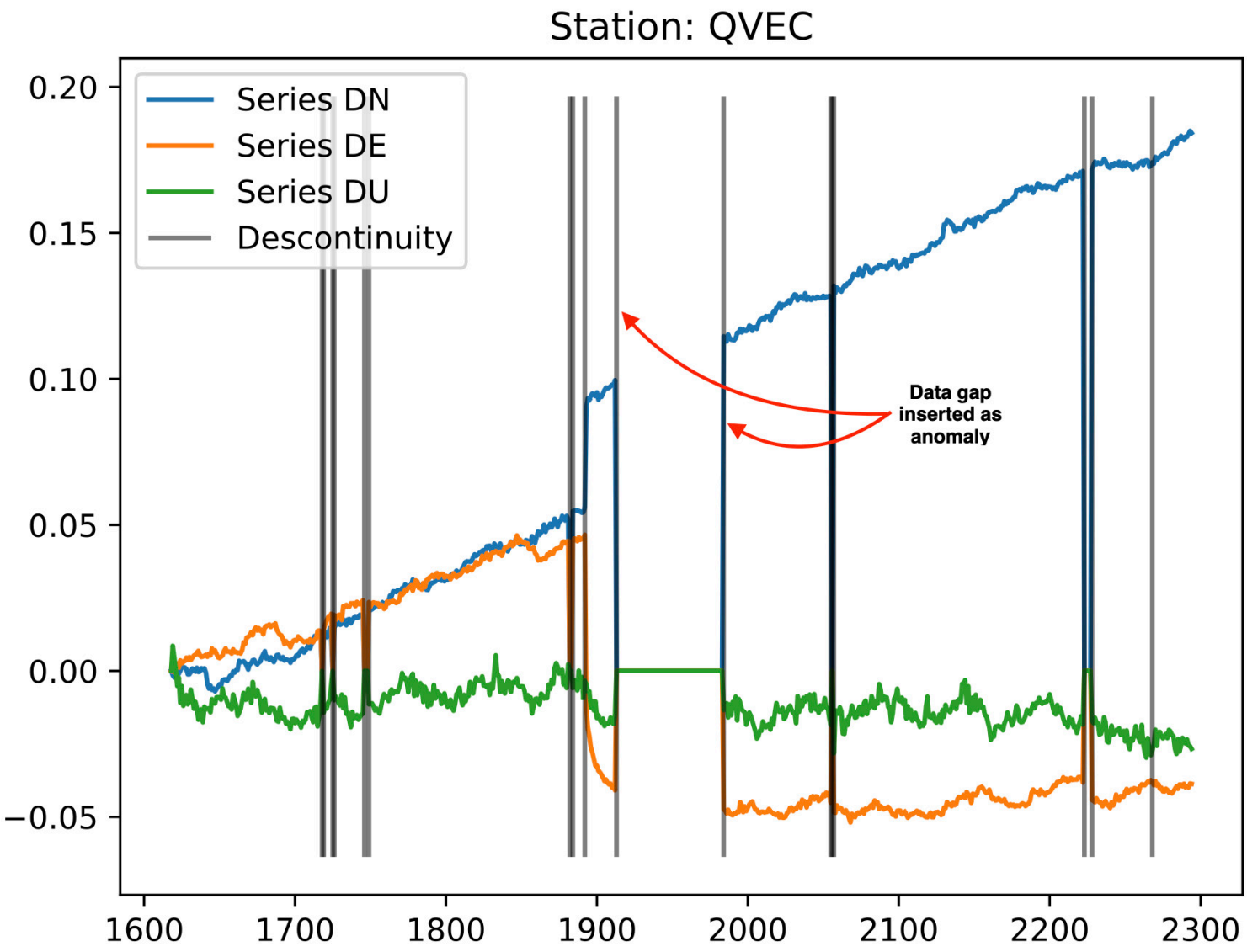


Figure 3

QVEC station showing an example of a data gap that was inserted as an anomaly, considering that the anomaly models usually demand a constant time frequency to operate.

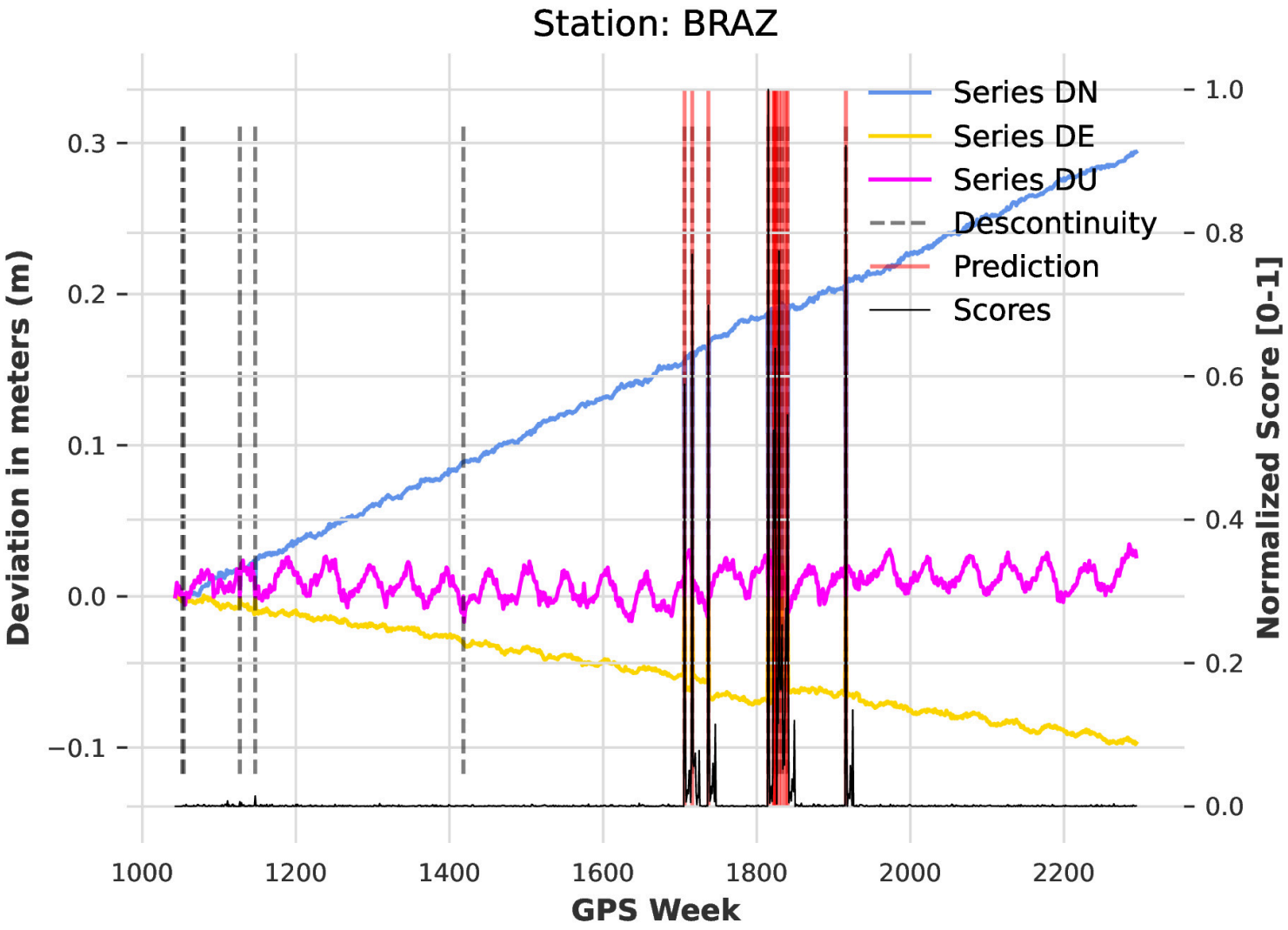


Figure 4

BRAZ station time series (DN, DE, and DU) and the registered discontinuities (vertical black dotted lines) overlayed with the predicted discontinuities (vertical red lines) and anomaly scores.

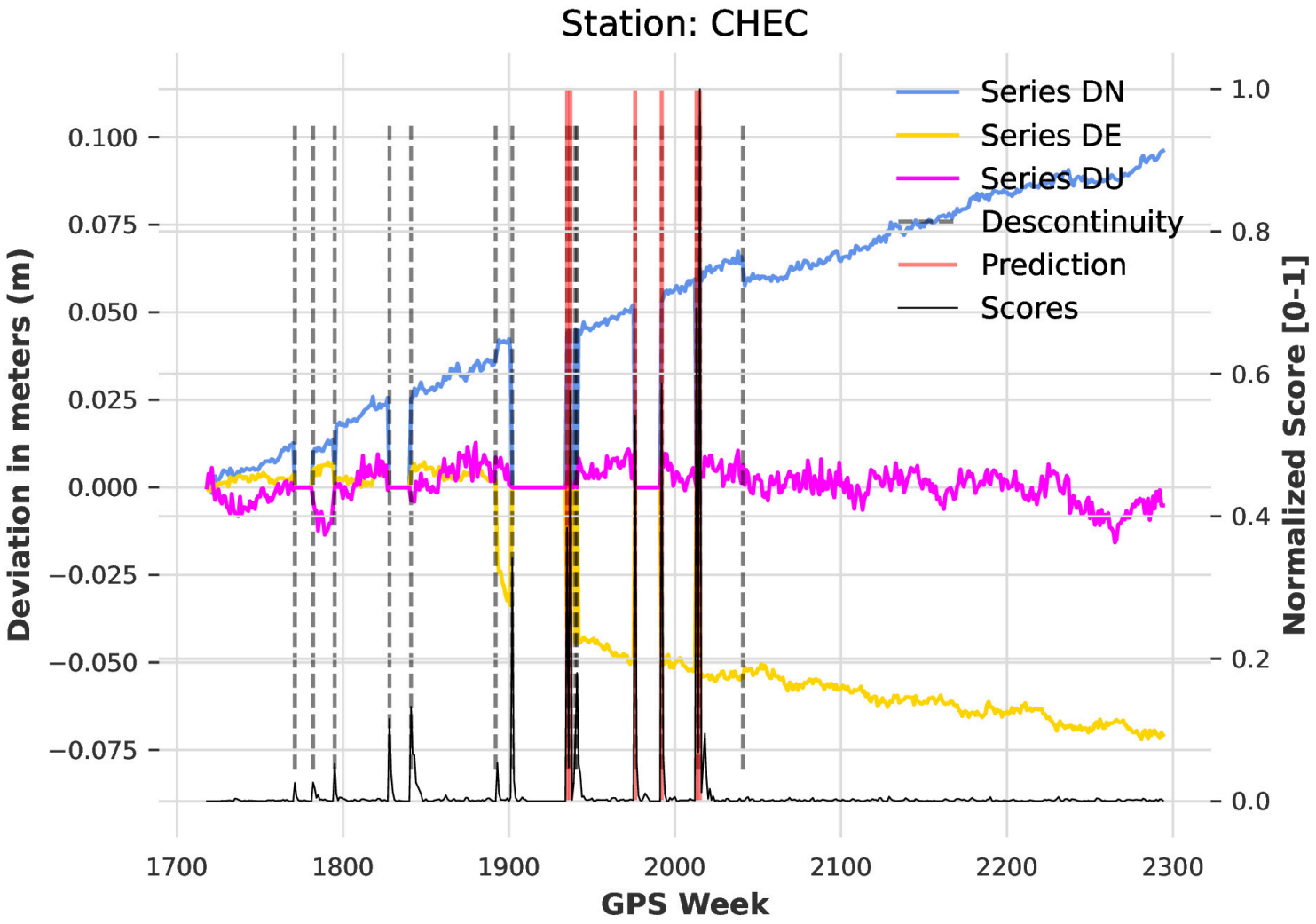


Figure 5

CHEC station time series (DN, DE, and DU) and the registered discontinuities (vertical black dotted lines) overlayed with the predicted discontinuities (vertical red lines) and anomaly scores.

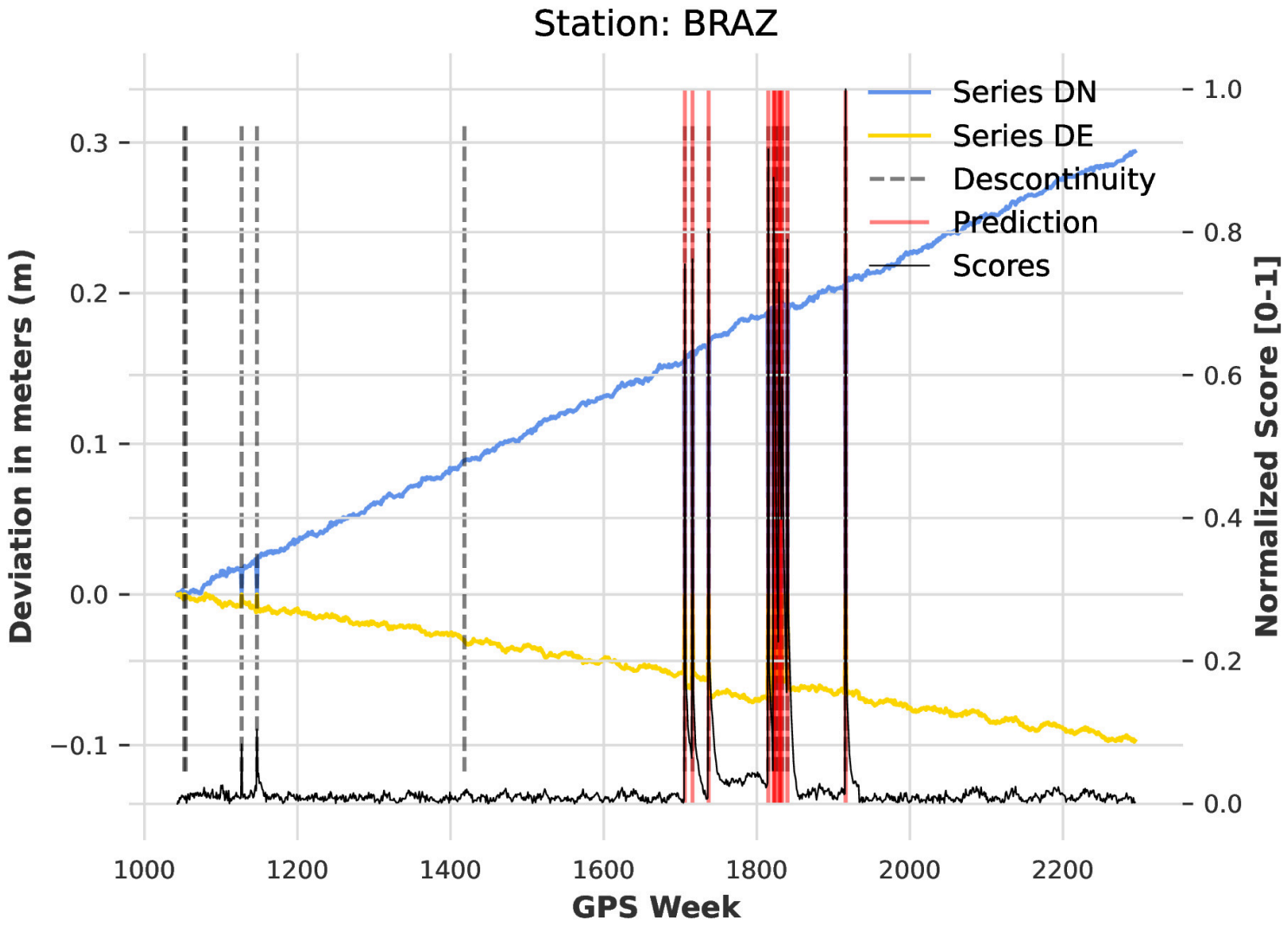


Figure 6

Kalman/Norm scorer results on the BRAZ dataset.

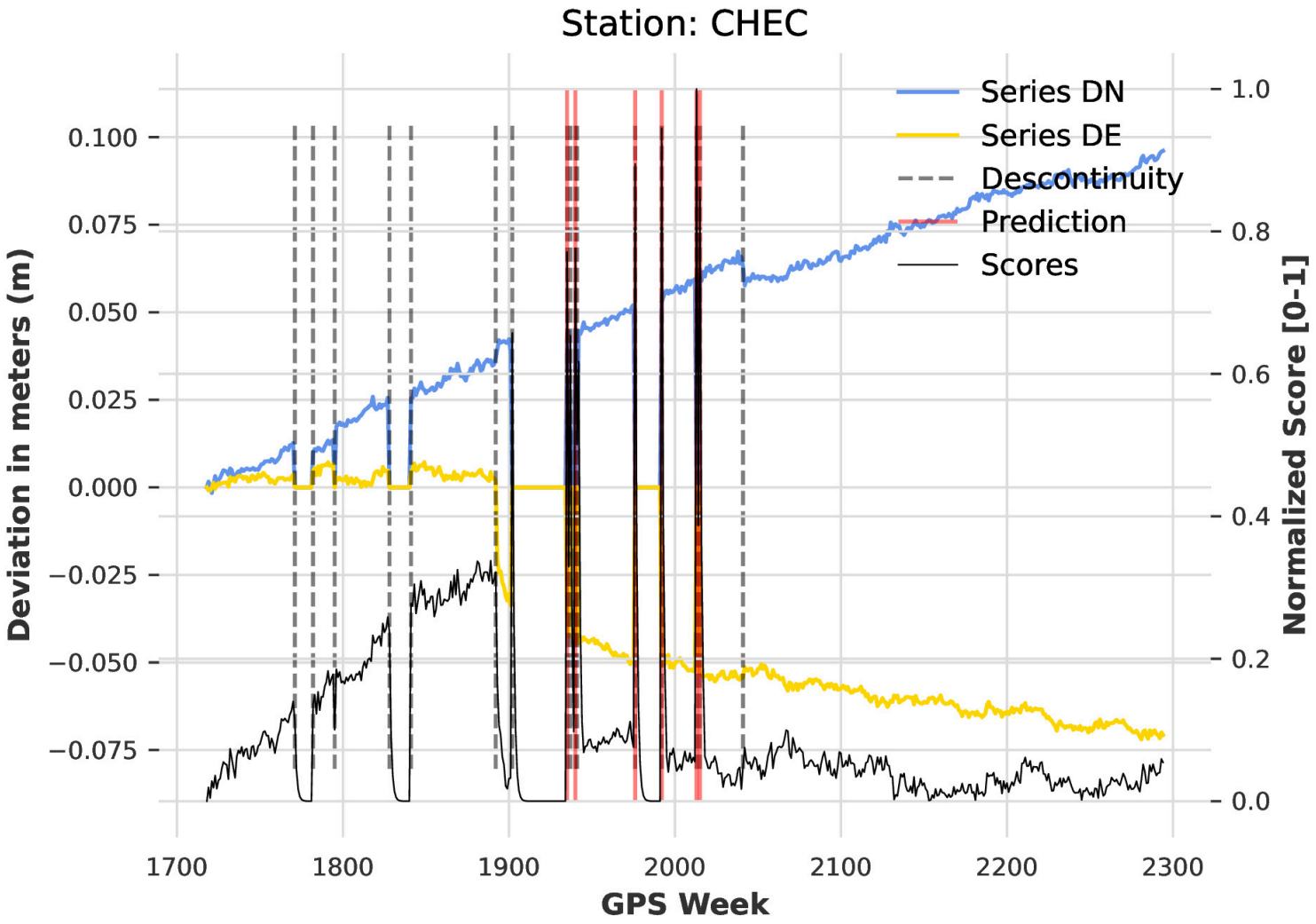


Figure 7

Kalman/Norm scorer results on the CHEC dataset.